Bench condition B: optimized layout, 20 LEDs (4 strings of 5), 22 mm minimum pitch, 21 mm gap, 3000 K, 55 mA. LED heat only (no Head electronics on the bench). Coordinates: reports/bench-B-optimized-20-led-coordinates.csv.

LeLamp Head: LED layout, light and heat

Generated 10/15/2026, 2:34:16 PM by Head LED Placement Studio · geometry from the Head STEP (gap 21 mm) · 20 LEDs at 55 mA, 3000 K, typical flux · dark room

Summary

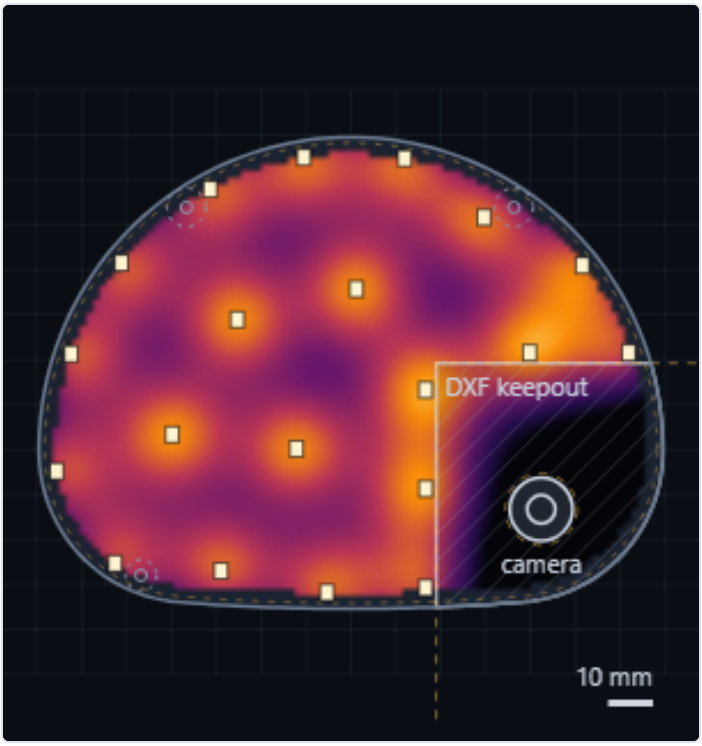
- **Light:** 234 lm out (target 300), desk minimum 291 lx at 40 cm (target 500), evenness 4% (target 65%; outside the keepout corner: 86%). Only 40% of the LED light leaves through the diffuser; 42% is absorbed by the walls, chamfer and lip.
- **Heat (worst case):** board 64 °C, LED chips 65 °C (datasheet limit 125), diffuser surface 44 °C, PLA bosses 63 °C (PLA softens ≈ 55 °C). 2.52 W in: LEDs 2.52 W + electronics 0.00 W.
- **Biggest passive lever:** Black back + aluminium standoffs + Al Head Back → board 36 °C (-28 °C).
- **Warm-up:** 95% of the temperature rise after 42 min; bench runs should last at least that long.
- **Fails:** Light out, Desk light (min), Evenness, Board temperature, PLA bosses touching the board, Diffuser outside (touch), Supply headroom.

Designs compared

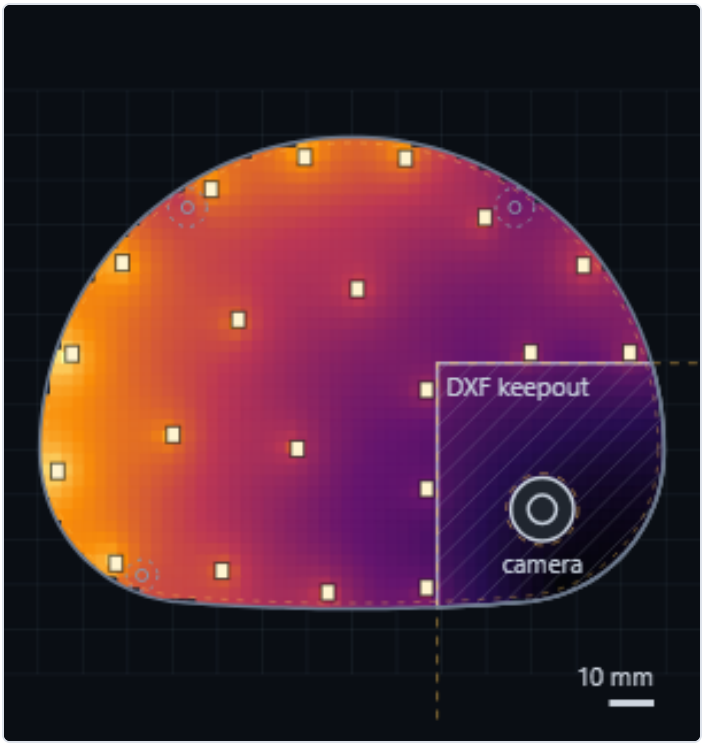
Design	Gap	Current	Evenness	Variation	Light out	Desk min / avg	Board / LED chip	PLA bosses	Diffuser surface	LED power	Light: out / walls / camera ring
This design	21 mm	55 mA	4% (outside the keepout corner 86%)	22.9%	234 lm	291 / 360 lx	64 / 65 °C	63 °C	44 °C	2.97 W	40% / 42% / 0.07%
Baseline: hex grid (20 LEDs)	21 mm	55 mA	2% (outside the keepout corner 35%)	35.6%	275 lm	338 / 423 lx	64 / 65 °C	63 °C	44 °C	2.97 W	47% / 32% / 0.05%

Every design is recomputed with the current model (heat: worst case). Targets: brief (300 lm, 65 %, 40 °C, 4 W, provisional); desk 500 lx at 40 cm (Ben's Notion, EN 12464-1); LED chip 125 °C (Cree Pro9 datasheet).

This design



Diffuser brightness (yellow bright, purple dark), with the LED layout.



Board temperature map (worst case): 62.3–63.8 °C.

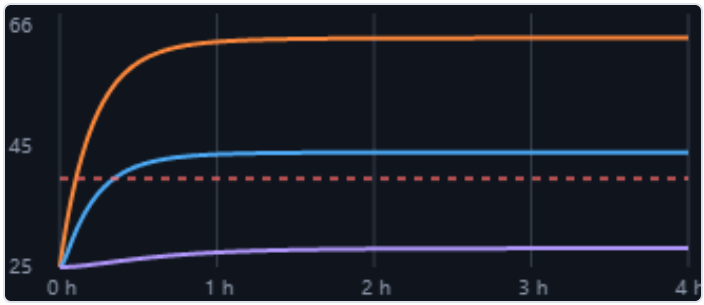
Where the light goes

Destination	Share	Lumens
Out through the diffuser	40%	234
Absorbed / scattered in the diffuser	17%	100
Walls, chamfer and lip (dark PLA)	42%	247
Camera barrel	0.5%	2.9
Escapes undiffused round the camera	0.07%	0.4

Heat

Quantity	Value
Heat in	2.52 W (LEDs 2.52 + electronics 0.00)
Leaves via	front 1.77 W · back cavity 0.62 W · bosses 0.13 W
Board hottest / coolest	63.8 / 62.3 °C
Hottest LED chip	65.4 °C (limit 125)
Diffuser outer surface (max)	44.1 °C
Head Back inside / outside	28.2 / 27.8 °C
Ben's network (LEDs only, front only)	76.5 °C

Warm-up after switch-on



Orange: hottest board point; blue: diffuser; purple: shell. 63 % of the rise after 13 min, 95 % after 42 min. Lumped 3-node model; heat capacities from STEP volumes × typical material values (assumed).

Passive cooling what-ifs (one change each)

Option	Board	Change	LED chip	PLA bosses
Board back painted / anodised black emissivity 0.1 → 0.9 (bare vs black-anodised aluminium, typical)	50 °C	-14 °C	51 °C	49 °C
Aluminium standoffs instead of PLA bosses boss k 0.13 → 138 W/m·K	63 °C	-1 °C	64 °C	62 °C
Head Back in PC instead of PLA wall k 0.13 → 0.2 W/m·K (Ben's PC value)	63 °C	-1 °C	65 °C	63 °C
Head Back in aluminium wall k → 138 W/m·K; outside emissivity 0.85 (anodised, typical)	61 °C	-3 °C	63 °C	60 °C
Electronics at typical power your table: 2.0 W instead of 4.6 W worst case	89 °C	+25 °C	90 °C	88 °C
Electronics off this board LED heat only (as in Ben's calc)	64 °C	0 °C	65 °C	63 °C
Black back + aluminium standoffs + Al Head Back the three passive changes together	36 °C	-28 °C	38 °C	34 °C

Which assumption matters most (one at a time)

Input	Range	Board low → high
Electronics power	your table: 2.0 typical – 4.6 W worst	89 → 116 °C
Air movement (Ben's cases)	Ben: Nu 1–3, h 2–25	47 → 64 °C
Board back emissivity	bare 0.05 – painted 0.9 (finish unknown)	66 → 50 °C
Room temperature	20–30 °C	60 → 68 °C
LED heat share k_h	Hui & Qin: 0.85; Qin & Hui: 85–90 %	62 → 66 °C
PLA conductivity	0.10–0.20 W/m·K (filament unknown)	64 → 63 °C
Dielectric k (chip only)	MCPCB 1–3 W/m·K	65 → 66 °C (chip)
Aluminium thickness	MCPCB 1.0–2.0 mm (stack-up unknown)	64 → 64 °C
Shell emissivity	0.80–0.95	64 → 64 °C

Constraint checklist

	Check	Now	Limit	Limit from
✗	Light out	234 lm	≥ 300 lm	Brief (provisional 300 lm)
✗	Desk light (min)	291 lx	≥ 500 lx	Ben's Notion target (EN 12464-1)
✗	Evenness	4 % whole face (outside the keepout corner 86 %)	≥ 65 %	Brief (provisional 65 %); scored over the whole lit face unless set to outside the keepout corner
✗	Board temperature	63.8 °C	≤ 40 °C	Brief (provisional 40°C goal)
✓	LED junction	65.4 °C	≤ 125 °C	Cree Pro9 datasheet p.4 (Tj max)
✗	PLA bosses touching the board	63.3 °C	≤ 55 °C	PLA glass transition ≈ 55–60 °C (textbook); bosses from the STEP
✓	Head Back inside (PLA)	28.2 °C	≤ 55 °C	PLA glass transition ≈ 55–60 °C (textbook)
✗	Diffuser outside (touch)	44.1 °C	≤ 40 °C	Brief's 40°C goal applied to the touch point it asks you to probe
✓	LED power	2.97 W	≤ 4 W	Brief / heat-calc doc (4 W reference)
✓	Placement	0 LED(s) break a limit	0	DXF outline, lens, screw holes, 2835 body size
✓	Driver current	55.0 mA	≤ 60.5 mA	R82 = 820 Ω (Head_Board.pdf); 41 × 1.21 V / R from the TLC59711 datasheet (not in the context files)
✓	LED current rating	55.0 mA	≤ 480 mA	Cree Pro9 datasheet p.4
✓	Ambient rating	25 °C	-40 to 105 °C	Cree Pro9 datasheet p.4
✗	Supply headroom	-0.13 V	≥ 0.4 V	LMR51420 divider (13.37 V) − 5 × Vf; TLC59711 regulation
✓	Driver channels	4 strings	≤ 12	TLC59711 (12 channels)

Assumptions and values not in the project files

Item	Value	Basis
Driver current limit	60.5 mA	R82 = 820 Ω from Head_Board.pdf × TLC59711 formula 41 × 1.21 V / R (datasheet not in the project files; its channel rating is 60 mA)
LED supply VDD_LED	13.37 V	100k/4.7k divider from Head_Board.pdf × LMR51420 0.6 V reference (datasheet not in the project files)
Regulation headroom limit	0.4 V	my engineering choice
Metal-core stack-up	1.6 mm Al k 138, 1 oz Cu, 0.1 mm dielectric k 2, back ε 0.1	typical values until the PCB stack-up exists
PLA properties	k 0.13, ε 0.9, softens ≈ 55–60 °C	typical values; check the filament datasheet
Electronics placement	spread evenly	positions unknown
FR4 comparison	in-plane k 0.8, 1.6 mm	typical laminate value (comparison only)
Heat capacities (warm-up)	Al 2700 kg/m³ × 900, PC 1200 × 1200, PLA 1240 × 1800 J/kg·K	typical material values; volumes from the STEP
What-if materials	black back ε 0.9, Al standoffs/shell k 138, PC k 0.2, anodised ε 0.85	typical values; PC k is Ben's value
Sensitivity ranges	e.g. back ε 0.05–0.9, PLA k 0.10–0.20, Al 1–2 mm, room 20–30 °C	typical spreads; air cases from Ben, power range from your table
Desk reflectance (Real look only)	0.5	display only
LED colour on screen	blackbody fit	Tanner Helland approximation (display only)
Optimizer settings, colour thresholds, clearances	e.g. penalty 50, 1.3 / 1.0 / 3.5 / 4.5 mm	my engineering choices (speed and safety margins, not physics)
Papers not in the project files	Su 2012, Du/Faber/Gunzburger 1999, Bridson 2007	cited for algorithms only (optimizer, Even out, Randomize); context/papers holds Moreno, Yang, Hui & Qin

All other values trace to the Cree Pro9 datasheet, the Head STEP, Head_Board.pdf, the brief, Ben's Notion pages, the cited papers, or your answers. Full list: the app's Evidence panel (step 5) / evidence.js.

Build record: LED positions

mm from the head's left / bottom edges (the dome centre is at x = 69.55, y = 35.47 in these units). Nearest-neighbour spacing 22.0–27.9 mm.

LED	String	Duty	x from left	y from bottom
1	4	100%	3.97	30.65
2	4	100%	16.98	10.13
3	4	100%	40.57	8.47
4	4	100%	86.19	4.71
5	3	100%	29.72	38.80
6	4	100%	64.20	3.66
7	3	100%	86.22	26.72
8	2	100%	7.18	56.76
9	2	100%	44.23	64.38
10	3	100%	57.41	35.64
11	2	100%	86.19	48.74
12	3	100%	109.37	57.02
13	3	100%	131.45	57.02
14	2	100%	18.39	77.02
15	1	100%	38.17	93.40
16	2	100%	70.76	71.19
17	1	100%	99.21	87.13
18	1	100%	121.14	76.46
19	1	100%	59.04	100.51
20	1	100%	81.51	100.27

Method and sources (short)

- Light: each LED a cos^m (m = 1 from the 120° beam) source onto the diffuser at the optical height (17.9 − 0.62 mm), Moreno 2006 Eq. 1 × incidence cosine = Ben's derivation; walls absorb; diffuser transmission 0.70.
- Desk: each diffuser patch a Lambertian source onto the desk plane (Ben's method). Evenness = darkest/average (brief); variation = CV (Yang 2009, Su 2012).
- Heat: 2-D board conduction + Ben's front network + back cavity (STEP depths) + PLA bosses with radial spreading into the shell; LED heat k_h = 0.85 (Hui & Qin 2009); reduces exactly to Ben's network (check 20).
- Checks: node checks/examples.js (26 checks against the sources) in the project repository.